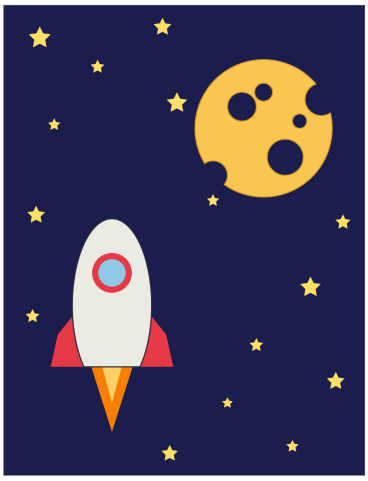
Mission to the Cheese Moon

The mission



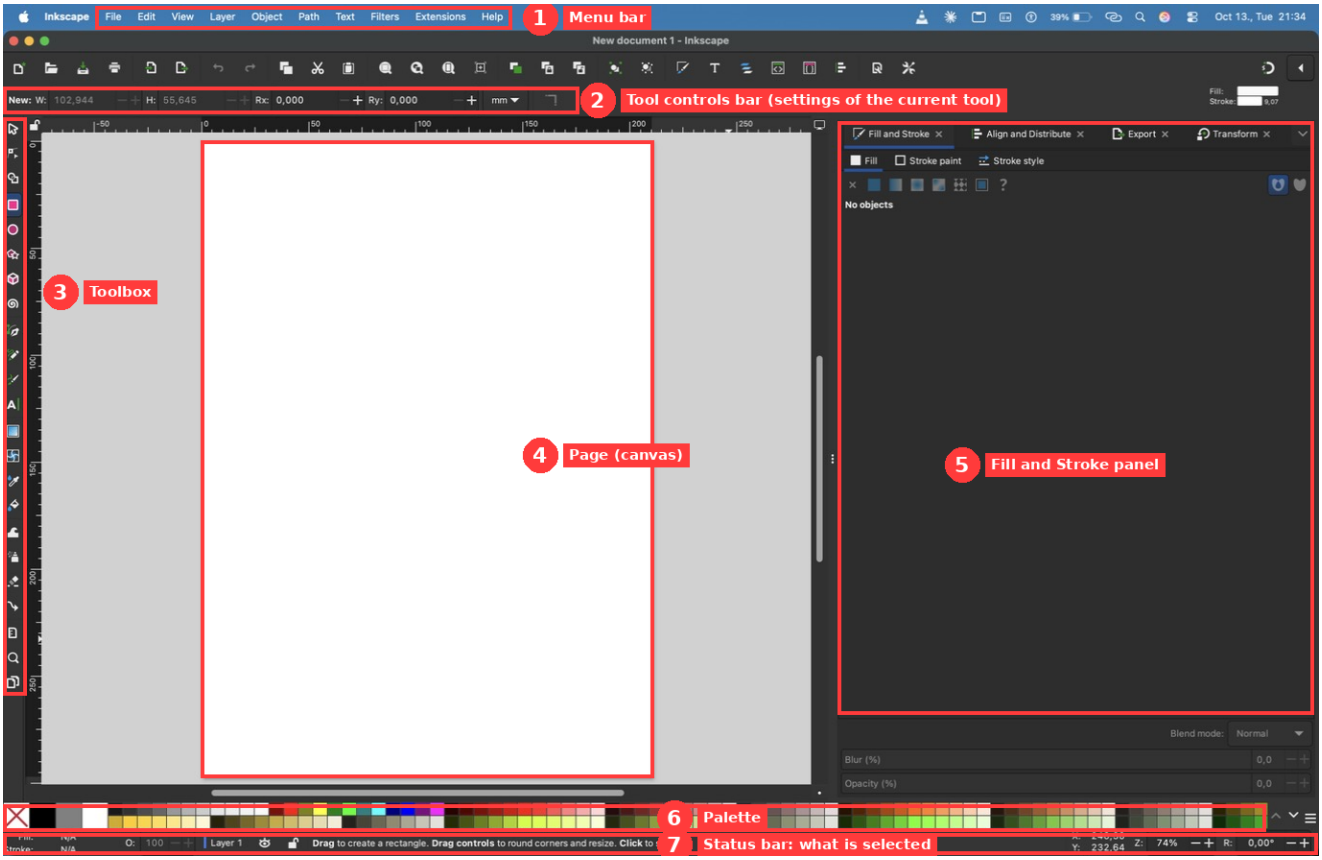
Legend says the Moon is made of cheese. The next space mission will find out – and you are designing the launch poster. You only need what you already know: shapes, fill and stroke, the Node tool, Union, Intersection and Difference.

No keyboard shortcuts – every command is in a menu or on a toolbar. Hand in an SVG and a PNG. Colours, sizes and the shape of the rocket are up to you; what counts is that every element is made with the operation in the table.

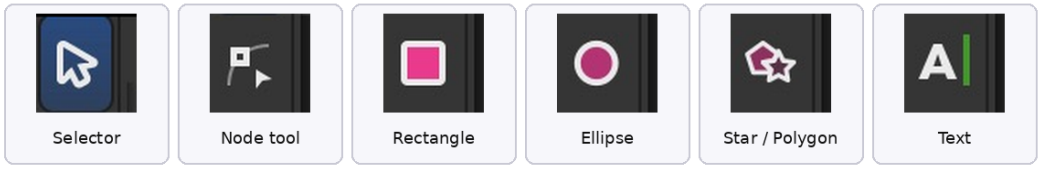
Element	How it is made	What you practise
Night sky and stars	Rectangle tool + Star tool	basic shapes, fill, stroke
Cheese Moon with craters	big circle, small circles → Union → Difference	Union, Difference
Rocket body	ellipse + rectangle → Intersection	Intersection
Fins	rectangle → path → nodes; duplicate, flip	Node tool
Porthole	two circles → Difference (ring)	Difference
Flame	polygon with 3 corners	basic shape, stacking order

Before you start

This is Inkscape. You will use the numbered parts: menu bar, tool controls bar, toolbox, page, Fill and Stroke panel, palette, status bar.



The tools you will use (toolbox on the left edge – hover an icon to see its name):



File ▸ New a new document

File ▸ Save As... surname_cheesemoon.svg – then File ▸ Save every few minutes

View ▸ Zoom ▸ Zoom Page shows the whole page

Three things to remember:

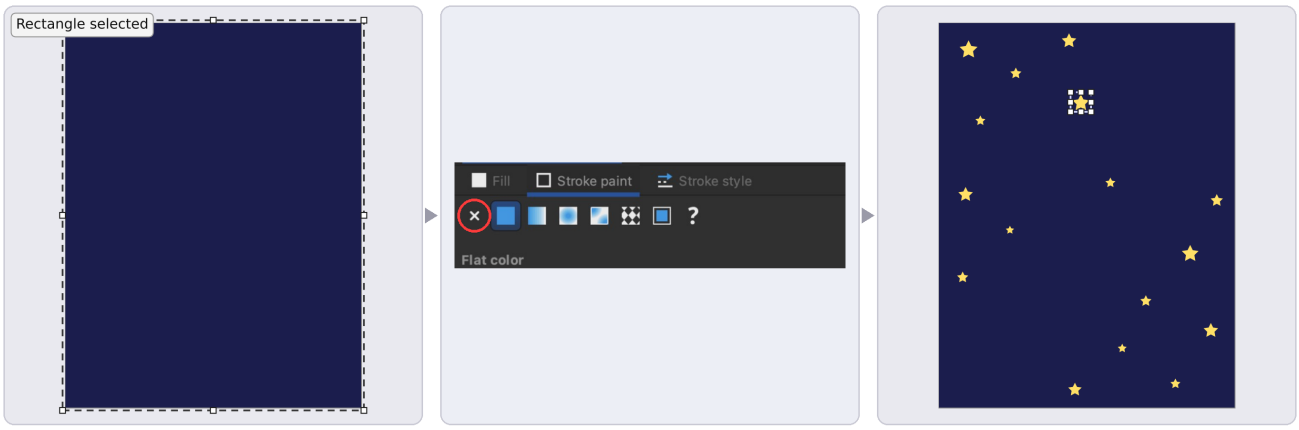
+ **Object ▸ Fill and Stroke...** palette swatch = fill colour; stroke colour and on/off in Fill and Stroke, Stroke paint tab

+ **Shift** select several objects: click the first, Shift + click the others (the status bar counts them)

Path ▸ Difference the object on TOP is cut out of the one underneath – exactly 2 objects selected

No keyboard shortcuts in this worksheet – every command is in a menu or on a toolbar. The caption under each picture tells you which tool, colour or menu to use.

Step 1 – Night sky and stars



1 → Drag over the whole page. Click dark blue in the palette.

2 **Object ▸ Fill and Stroke…** Stroke paint tab ▸ click the X = no stroke.

3 Star mode, Corners 5. Drag out about 8 stars. Click yellow.

Leave the top-right corner empty (Moon) and the bottom-left area empty (rocket).

Step 2 – The Cheese Moon

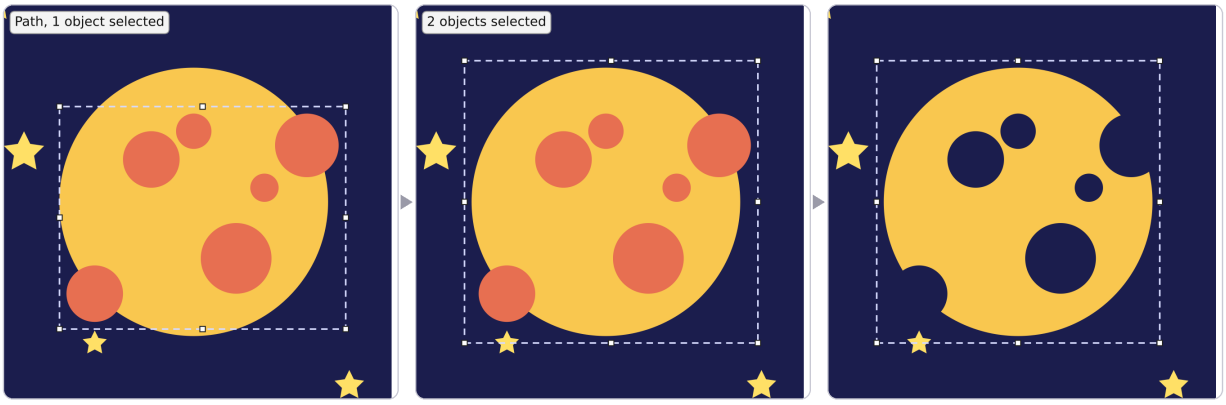
Union first (the craters become one object), then Difference (the craters are cut out of the Moon).



1 → Big circle, top right. Click yellow.

2 5–6 small circles on it, 2 hanging over the edge.

3 + **Shift** Click one small circle, Shift + click the others.



4 **Path ▸ Union** The craters are now ONE object.

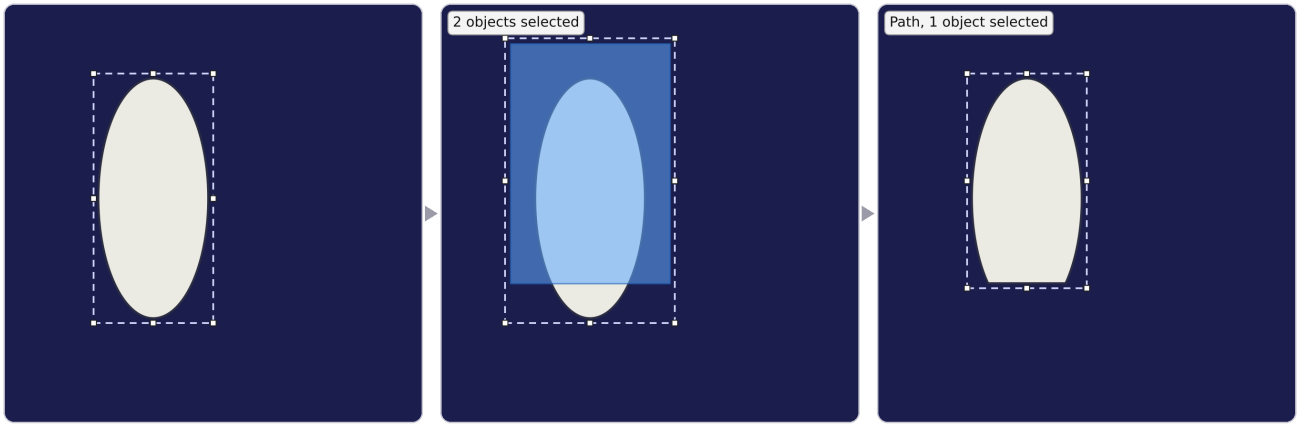
5 + **Shift** Click the Moon, Shift + click the craters.

6 **Path ▸ Difference** Holes and a bitten edge: the Cheese Moon.

If the Moon disappeared instead of the craters: Edit ▸ Undo, select the craters, Object ▸ Raise to Top, try again.

Step 3 – Rocket body

Intersection keeps only the part the two shapes share.



1 → +

Tall ellipse, bottom left. Fill white; Stroke paint tab ▸ a dark colour.

2 → + **Shift**

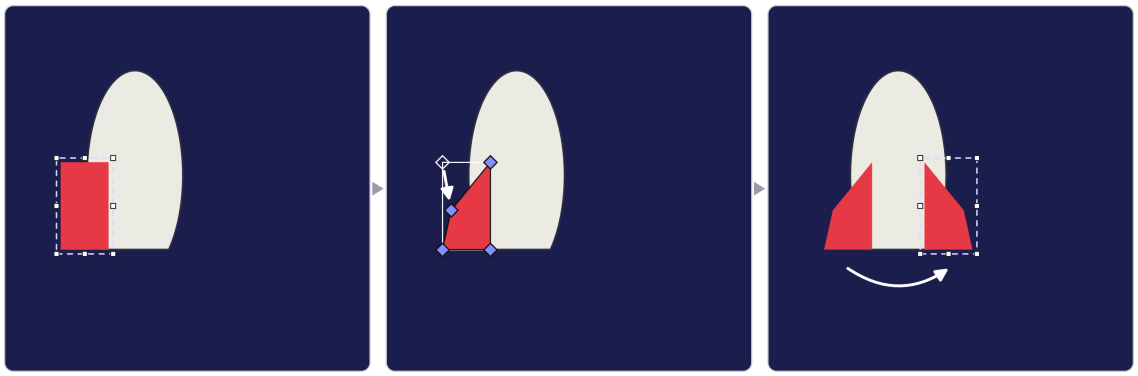
Rectangle over the TOP part. Click one, Shift + click the other.

3 **Path ▸ Intersection**

Only the shared part stays: a flat-bottomed body.

Step 4 – Fins with the Node tool

A rectangle becomes a fin by pulling one corner; the other fin is a flipped copy.



1 →

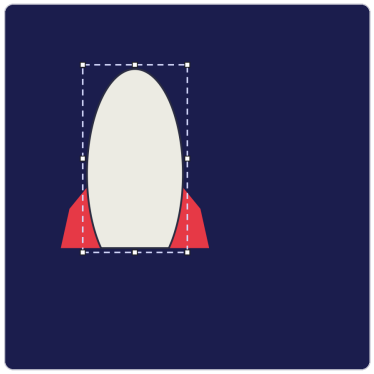
Red rectangle at the bottom left, overlapping the body.

2 **Path ▸ Object to Path** →

Node tool: drag the top-left corner node down = the fin.

3 **Edit ▸ Duplicate** →
Object ▸ Flip Horizontal

Then drag the copy to the right side of the body.



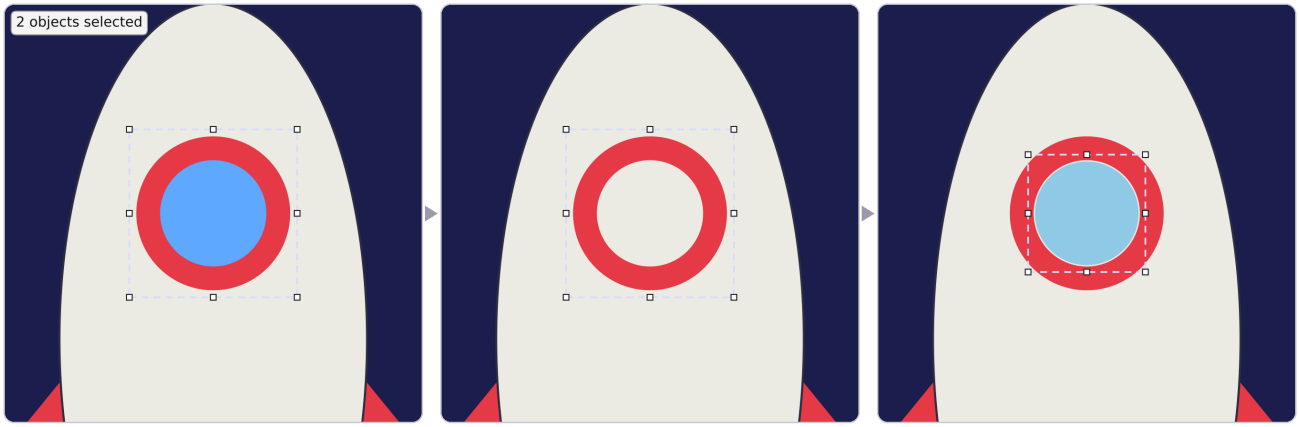
4 → **Object ▸ Raise to Top**

Click the body: it moves in front of the fins.

If the whole shape moves when you drag a node, you are still on the Selector – pick the Node tool from the toolbox.

Step 5 – Porthole

A ring = Difference of two circles.



1 → →

Red circle on the body, then a smaller circle inside it. Select both.

2 **Path ▸ Difference**

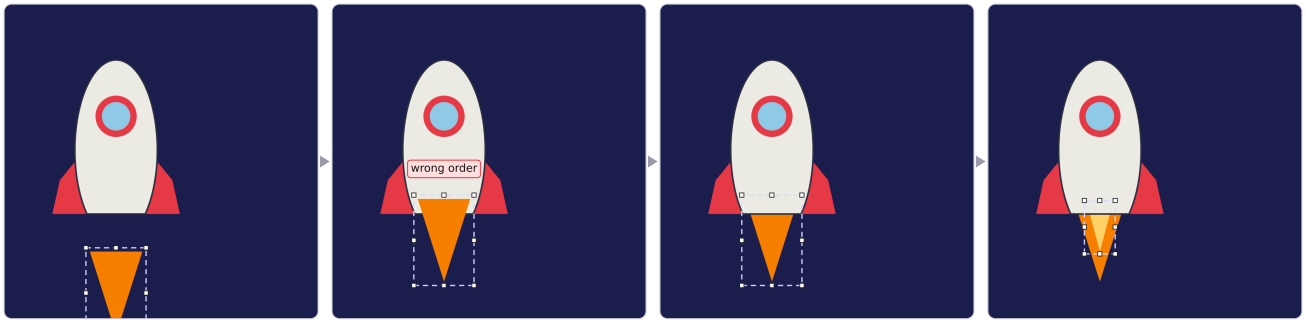
The small circle cuts the hole: a ring.

3 →

Light-blue circle in the hole = the glass.

Step 6 – Flame and stacking order

The flame must go behind the body: Object ▸ Lower moves it one step back each time. Then a yellow inner flame on top.



1 New: Corners: 3

Polygon mode, Corners 3. Drag DOWN from the centre.

2 →

Move it under the rocket, overlapping. Click orange.

3 **Object ▸ Lower**

Repeat until the flame is BEHIND the body.

4 **Edit ▸ Duplicate** →

Inner flame: make the copy smaller, click yellow.

Object ▸ Lower to Bottom would hide the flame behind the sky – then Object ▸ Raise. Finally View ▸ Zoom ▸ Zoom Page to see the whole poster.

Step 7 – Save and hand in

File ▸ Save the SVG file

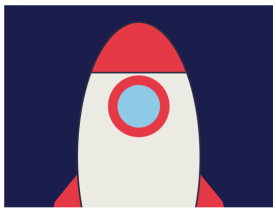
File ▸ Export... choose Page, name it surname_cheesemoon.png, click Export

Hand in both files: `surname_cheesemoon.svg` and `surname_cheesemoon.png`. (Older Inkscape: File ▸ Export PNG Image...)

Bonus – if you have time left

Pick one or more. Each one uses the same operations as the main task.

Task	Recipe
Red nose cone	An ellipse the size of the body + a rectangle over the TOP part → Path ▸ Intersection → red fill, place it on the tip of the body
Shooting star	A star + three long thin triangles fanning out behind it, each starting inside the star (Polygon mode, 3 corners, stretched) → select all → Path ▸ Union
Ringed planet	A circle + a flat ellipse with a smaller flat ellipse on top → Path ▸ Difference → put the ring in front of the planet
Crater shadows	Select the finished Moon → Edit ▸ Duplicate → fill the copy darker orange → Object ▸ Lower (behind the Moon) → move it a little: the craters get shadows
Poster title	Text tool (toolbox), click at the top of the page, type MISSION TO THE CHEESE MOON! – big bold letters, yellow
Tilted rocket	Drag a selection box around the whole rocket, Object ▸ Group, click it a second time and rotate it with the corner arrows



Nose cone: ellipse + rectangle over the TOP ▸ Path ▸ Intersection, red fill



Shooting star: star + 3 thin triangles ▸ Path ▸ Union



Ringed planet: flat ellipse − smaller ellipse ▸ Path ▸ Difference, placed in front of the planet



Crater shadows: Edit ▸ Duplicate the Moon, darker orange fill, Object ▸ Lower (behind the Moon)

Bonus ideas: nose cone, shooting star, ringed planet, crater shadows

Checklist before handing in

- ☐ The sky covers the whole page
- ☐ At least 8 stars
- ☐ At least 5 craters on the Moon, two of them biting into the edge (Difference)
- ☐ The rocket body has a straight bottom edge (Intersection)
- ☐ Two mirrored fins, with the body in front of them (Node tool + flip)
- ☐ A ring-shaped porthole with glass (Difference)
- ☐ The flame is behind the body and has a yellow inner flame (stacking order)
- ☐ The rocket body has a dark stroke, the sky has none
- ☐ Both SVG and PNG saved, with your surname in the file name